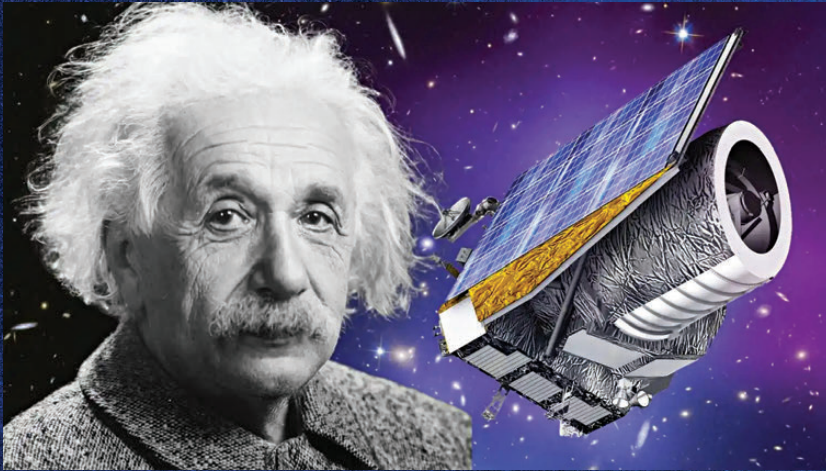




Einstein Rings and the cosmic lens

How NGC 6505 is reshaping our understanding of gravity

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The universe has a habit of surprising us, revealing its deepest secrets in the most unexpected ways. One such revelation arrived in the form of an astonishing discovery around the elliptical galaxy NGC 6505 – a complete Einstein ring, captured by the keen eyes of the Euclid space telescope. This is no ordinary find. It is a cosmic phenomenon that elegantly showcases the warping of space-time, a consequence of gravity predicted over a century ago by Albert Einstein. Yet, beyond its scientific allure, this discovery holds

implications that stretch far beyond the domain of astrophysics. It forces us to reconsider our understanding of dark matter, the fabric of the universe, and even the potential for intergalactic observations that could redefine modern cosmology.

What exactly is an Einstein Ring?

To appreciate the magnitude of this discovery, one must first understand what an Einstein ring is. In the grand cosmic theatre, light from a distant galaxy does not always travel in a straight line. When a massive foreground galaxy, like NGC 6505,

sits directly between us and a more distant source, its gravity acts as a lens, bending the light around it. If the alignment is precise, the light forms a near-perfect circle around the intervening galaxy, creating what astronomers call an Einstein ring.

These rings are more than just celestial eye candy. They serve as powerful tools for measuring the mass distribution of galaxies, allowing astronomers to peer into regions otherwise obscured by luminous matter. More crucially, they offer rare glimpses into the enigmatic realm of dark matter, the unseen substance that makes up approximately 85% of the universe's mass.

A discovery decades in the making

The presence of a strong gravitational lens around NGC 6505 was, until now, completely unknown. This elliptical galaxy has been observed in various wavelengths for decades – in X-rays, radio, and optical imaging – but its role as a cosmic lens remained undetected. It was only when Euclid turned its gaze toward this part of the sky that the complete Einstein ring became evident, marking the first discovery of its kind in an NGC object. The data collected from Euclid's Visible Camera (VIS) and Near-Infrared Spectrometer and Photometer (NISF) provided unparalleled clarity, enabling researchers to measure the source galaxy's redshift at $z = 0.406$ and confirm the lensing effect. The Keck Cosmic Web Imager further bolstered the findings, revealing the remarkable precision with which this natural gravitational lens bends light across billions of years. **d**

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